Lecture Notes

on Sample Spaces, Events, and Probability

by Arnaja Mitra

1 Probability of Events

Example: Suppose a coin is flipped once. What are all possible outcomes?
Let 'H' denote *head* and 'T' denote *tail*. So, if someone flips a coin only once, there are two possible outcomes, which we can write as a set:

$$S = \{H, T\}.$$

1.1 Sample Space

The complete list of possible outcomes for an experiment $\mathcal{E}$ is called the *sample space* for $\mathcal{E}$, and in general we denote it by S. The elements of the set S are called *elementary events*.

In general, if

$$S = \{e_1, e_2, \ldots, e_n\},$$

then each e_i ($i \in \{1, 2, \ldots, n\}$) is an elementary event.

Example: If someone rolls a die, then

$$S = \{1, 2, 3, 4, 5, 6\}.$$

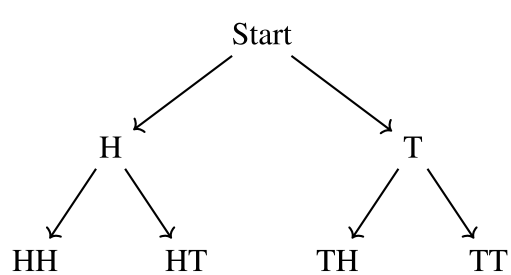
1.2 Event

Given a sample space S, a subset of S, say E (i.e., $E \subseteq S$), is called an *event*.

Note: E may contain none, some, or all elements of S.

Example Let a coin be flipped twice. What elementary events are in the event

$$E = \text{“first and second flips do not match.”}$$



So, the sample space for the given experiment "flipping a coin twice" is

$$S = \{HH, HT, TH, TT\}.$$

Hence, the event E is

$$E = \{HT, TH\}.$$

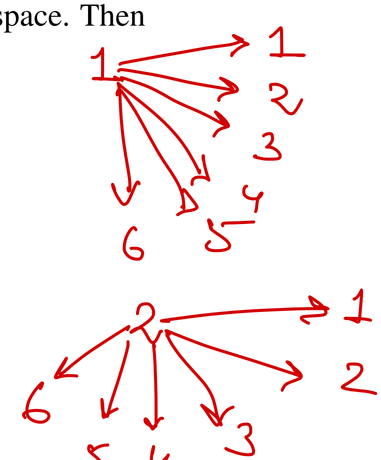
Example Let a single die be rolled. What is the sample space?
Suppose two dice are rolled. What is the sample space?
What elementary events are in the event "dice sum to 6"?

Let $\mathcal{E}_1$ be the experiment of rolling one die and S_1 be the corresponding sample space. Then

$$S_1 = \{1, 2, 3, 4, 5, 6\}.$$

Let $\mathcal{E}_2$ be the experiment of rolling two dice and S_2 be the corresponding sample space. Then

$$S_2 = \begin{array}{l} \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6), \\ (2,1),(2,2),(2,3),(2,4),(2,5),(2,6), \\ (3,1),(3,2),(3,3),(3,4),(3,5),(3,6), \\ (4,1),(4,2),(4,3),(4,4),(4,5),(4,6), \\ (5,1),(5,2),(5,3),(5,4),(5,5),(5,6), \\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}. \end{array}$$



Size of S_2: By the multiplication principle,

$$|S_2| = 6 \times 6 = 36.$$

Let E denote the event "dice sum to 6." Then

1+5 2+4 3+3

$$E = \{(1,5), (2,4), (3,3), (4,2), (5,1)\}.$$

4+2 5+1

2 Multiplication Principle

If action$_1$ in an experiment $\mathcal{E}_1$ can be performed in m ways and action$_2$ in experiment $\mathcal{E}_2$ can be performed in n ways, then a new experiment $\widehat{\mathcal{E}}$ consisting of action$_1$ followed by action$_2$ can be performed in $m \times n$ ways. Thus, the sample space of $\widehat{\mathcal{E}}$ contains mn elementary events.

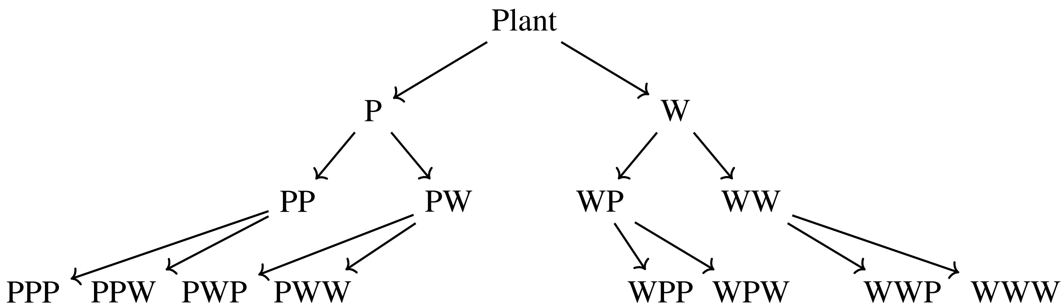
Note: It can be extended to as many actions as you want.

Explanation:

$$\begin{array}{ccc} \mathcal{E}_1 & & \mathcal{E}_2 \\ \text{way 1} & \longrightarrow & \text{way 1, way 2, \ldots, way n} \\ \text{way 2} & \longrightarrow & \text{way 1, way 2, \ldots, way n} \\ \vdots & & \vdots \\ \text{way m} & \longrightarrow & \text{way 1, way 2, \ldots, way n} \end{array}$$

Hence, total number of outcomes $= m \times n$.

Example: Let you have a plant that gives either *purple (P)* or *white (W)* flowers. If you have 3 seeds to be planted, what would the sample space of all possible outcomes be? List the elements in the event that there are *exactly two seeds that produce plants with purple flowers.*



(For three seeds where each outcome is a sequence of P's and W's:)

$$S = \{PPP, PPW, PWP, PWW, WPP, WPW, WWP, WWW\}.$$

The event that exactly two seeds produce purple flowers is

$$E = \{PPW, PWP, WPP\}.$$

3 Probability

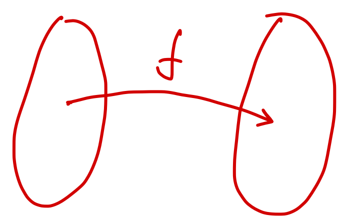
Let experiment $\mathcal{E}$ have sample space

$$S = \{e_1, e_2, \ldots, e_n\}.$$

The probability of each elementary event e_i in S is given by

p_i

$$\omega(e_i) = p_i,$$



where

$$\omega : S \longrightarrow [0, 1],$$

and ω is a function (called a **probability function**) satisfying:

1. $0 \leq p_i \leq 1$,

2. $\displaystyle\sum_{i=1}^{n} p_i = p_1 + p_2 + \cdots + p_n = 1.$

3.1 Probability of an Event

If $E \subseteq S$ is an event, then the probability of E is denoted as

$$E = \{e_1, e_2, \dots, e_k\} \subseteq S = \{e_1, e_2, \dots, e_n\}, \quad k \leq n,$$

and

$$P(E) = p_1 + p_2 + \cdots + p_k = \omega(e_1) + \omega(e_2) + \cdots + \omega(e_k).$$

i.e.,

$$P(E) = \sum_{e_i \in E} \omega(e_i).$$

Special Case (Equiprobable Events)

If all elementary events of S are *equally likely (equiprobable)*, then

$$P(E) = \frac{\text{\# of elements in } E}{\text{\# of elements in } S}.$$

If each elementary event in S has equal probability, then the sample space S is called a **uniform** or **equiprobable** sample space.

Example 1 (Biased Coin)

$$S = \{H, T\}, \qquad \omega(H) = \tfrac{1}{3}, \quad \omega(T) = \tfrac{2}{3}.$$
$$E = \{\text{getting } H\}, \quad \Rightarrow \quad P(E) = \omega(H) = \tfrac{1}{3}.$$

If the coin is not biased, then $\omega(H) = \omega(T) = \frac{1}{2}$.

$S = \{H, T\}$
$E = \{H\}$
$P(E) = \frac{1}{2}$

Example 2 (Continuation of the Last Example)

What is the probability that exactly two out of the three plants will produce purple flowers?

$$P(E) = \frac{3}{8}.$$

$P(S) = \frac{8}{8} = 1$

E_1 = having red flower

$P(E_1) = \frac{0}{8} = 0.$

4 Quick Notes on Basic Genetics Terms

1. **Gene:** A segment of DNA that controls a specific trait or characteristic.
   Example: The gene that determines eye color.

2. **Allele:** Different forms of the same gene that produce variations in a trait.
   Example: Alleles for eye color may be brown (B) or blue (b).

3. **Genotype:** The genetic makeup of an organism, its combination of alleles for a particular trait.
   Example: BB, Bb, or bb.

4. **Phenotype:** The observable physical or physiological trait expressed from the genotype.
   Example: Brown eyes (phenotype) result from genotypes BB or Bb.

5. **Punnett Square:** A diagram used to predict possible genotypes of offspring from parental crosses.
   Example: Crossing $Bb \times Bb$ gives genotypes BB, Bb, Bb, and bb.

Punnett Square Example: $Bb \times Bb$

	B	b
B	BB	Bb
b	Bb	bb

Genotypes: BB, Bb, bb; **Phenotypes:** 3 brown : 1 blue

6. **Carriers of Recessive Alleles:** Individuals who have one dominant and one recessive allele; they do not show the recessive trait but can pass it to offspring.
   Example: A person with genotype Bb carries the allele for blue eyes but has brown eyes.

Example A father and mother have blood types AO and AB, respectively. What is the probability that their child has blood type (a) A, (b) B, (c) AB, and (d) O?

Punnett Square

Father\Mother	A	B
A	AA	AB
0	OA	OB

Type (Phenotype)	Genotype(s)
Type A	AA, AO, OA
Type B	BB, BO, OB
Type AB	AB, BA
Type O	OO

Let E_1, E_2, E_3, and E_4 denote the events that their child has blood of type A, B, AB, and O respectively. Then,

$$\begin{aligned}
E_1 &= \{AA, OA\}, && \Rightarrow P(E_1) = \frac{2}{4}, \\
E_2 &= \{OB\}, && \Rightarrow P(E_2) = \frac{1}{4}, \\
E_3 &= \{AB\}, && \Rightarrow P(E_3) = \frac{1}{4}, \\
E_4 &= \{\varnothing\}, && \Rightarrow P(E_4) = \frac{0}{4} = 0.
\end{aligned}$$

Hence, the probabilities are:

$$\begin{aligned}
&\text{Type A:} && \frac{2}{4} = \frac{1}{2}, \\
&\text{Type B:} && \frac{1}{4}, \\
&\text{Type AB:} && \frac{1}{4}, \\
&\text{Type O:} && 0.
\end{aligned}$$